

REDROB · DATA & AI CHALLENGE

Ranking candidates the way a great recruiter would

A hybrid, fully-offline candidate ranker for the Senior AI Engineer role — built to read profiles, not keywords.

100,000 candidate pool **Top 100** shortlist **~1 min** CPU · no network

THE PROBLEM

Keyword filters can't see what matters



Keyword-stuffers

A “Marketing Manager” whose skills list is packed with RAG, Pinecone, embeddings. Perfect on keywords — wrong person.



Plain-language strengths

Built a recommendation system at a product company, but never wrote “vector DB.” Easy to miss, genuinely strong.



~80 honeypots

Subtly impossible profiles: 8 years at a 3-year-old company; “expert” in 10 skills with 0 months used. Forced to tier 0.



Behavioral twins

Identical on paper — separated only by who actually responds to recruiters and is still active.

A pure embedding or keyword system falls for all four. The JD even says so explicitly.

THE APPROACH

An explainable hybrid, not a black box

1 • Structured scoring

Seven explainable features per candidate: title fit, trust-weighted skills, career evidence, semantic match, experience band, domain, location.

2 • Modifiers

Two multipliers on the merit score: explicit JD disqualifiers (red flags) and behavioral availability (response rate, recency, open-to-work).

3 • Honeypot guard

Internal-consistency checks collapse impossible profiles beneath every genuine candidate — no ID special-casing.

Final score = weighted merit × red-flag penalty × behavioral modifier

Title fit + skill-trust defeat keyword-stuffers. **Career-description text** surfaces plain-language strengths. **Behavioral signals** encode who's actually hireable.

SCORING COMPONENTS

What each signal captures

COMPONENT	WEIGHT	WHAT IT CAPTURES	WHY IT MATTERS
title_fit	0.24	Is the title actually an ML/AI/SW engineer?	Decisive vs keyword-stuffers
skills_fit	0.18	Skills × trust (endorsements + duration + assessment)	Kills “expert, 0 months used”
career_evidence	0.18	Product-company + shipped ranking / search / recsys	Finds real builders
semantic	0.16	TF-IDF cosine: profile ↔ JD-derived query	Plain-language strengths
experience_fit	0.12	Alignment with the 5–9y band (ideal 6–8)	JD’s soft band
domain_fit	0.07	NLP / IR vs CV / speech / robotics	JD-named negative
location_fit	0.05	Indian metros / willing to relocate	No visa sponsorship

Weights live in config.yaml • the structured JD reading lives in role_spec.yaml

Reading profiles, not embeddings

Honeypot detection — internal consistency

Duration vs date span —

a role can't last 96 months across a 36-month window.

Tenure vs experience —

can't have worked more years than you've existed professionally.

Proficiency vs usage —

"expert" in many skills with 0 months used is impossible.

Skill vs career length —

a skill used longer than the whole career.

The keyword-stuffer test

"Marketing Manager" + 9 AI skills

the bad baseline ranks this #1.

Our title_fit scores it ~0.05. The skill-trust multiplier strips "expert" skills with no endorsements or usage. It lands near the bottom — where a recruiter would put it.

We never special-case IDs — detection generalizes and is defensible at interview.

RESULTS ON THE FULL POOL

100,000 candidates → a shortlist you can trust

0

honeypots in the top 100

0

keyword-stuffer roles in top 100

92/100

inside the 5–9y band (mean 6.6y)

~1 min

runtime · CPU · no network

14



AI Engineer

12



ML Engineer

10



Applied ML

8



Recsys Eng.

7



Senior DS

7



Search Eng.

6



NLP Eng.

4



Applied Sci.

EVERY RANK IS EXPLAINED

Reasoning a human can audit

#1

Lead AI Engineer with 6.7 yrs; strong role and skills alignment; relevant skills: Information Retrieval, Learning to Rank, Elasticsearch; responsive to recruiters (73%).

#10

Current role is Project Manager (14.5 yrs); core engineering-title fit is weak despite listed skills; not active in 218d.

bottom

Flagged as likely honeypot (role duration exceeds its date span — impossible tenure); ranked at the bottom despite surface keyword matches.

Grounded only in facts in the profile • names honest concerns • tone tracks the rank — built for Stage-4 manual review.

BUILT FOR PRODUCTION CONSTRAINTS

Reproducible, offline, within budget



CPU only

No GPU anywhere in the ranking path.



No network

Zero hosted-LLM / API calls during ranking.



< 5 min • < 16 GB

~1 min for 100k on a laptop CPU.



One command

```
python rank.py --candidates ... --out ...
```

Default semantic backend is TF-IDF — deterministic, no model download. An optional dense-embedding precompute is available; ranking stays offline either way.

WHAT WE BUILT

A ranker that reasons about fit

Explainable hybrid scoring

- title, skill-trust, career evidence, semantic, behavioral, honeypot guard.

Defeats the dataset's traps

- keyword-stuffers down, plain-language strengths up, honeypots out.

Production-ready

- offline, CPU, one command, ~1 min for 100k candidates.

Make hiring smarter.